

The open AI alternative

How ML Engineers Build It

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Content

- What's HF?
- Open vs Closed AI
- Does open AI makes sense?
- The role of an ML Engineer at HF
- Robotics at HF
- Building AI with HF tools



Hi, I'm Sergio 🙌

- Machine Learning Engineer @ Hugging Face
- PhD in AI
- Open source and AI
 - GSoC (JdeRobot)
 - ML Circle
- ◆ Teaching and Sharing



Sergio Paniego Blanco (@sergiopaniego)



What's Hugging Face? 🤗

AI open platform for building, sharing and deploying AI (HF Hub):

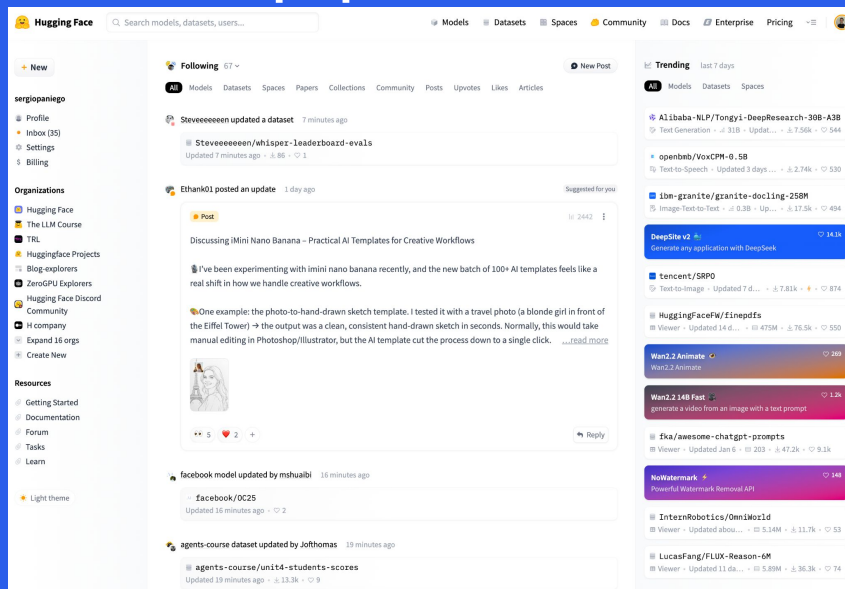
- Models, datasets, apps...
- Posts, blogs, courses, papers, leaderboards...

2.1M

public models

500K+

public datasets



1M+

daily downloads

1M+

daily visitors

What's Hugging Face? 🤗

Set of open source projects to empower AI development and advancement



Transformers



Datasets



Trackio



Hub



30+

Libraries



Transformers.js



Diffusers

timm



Accelerate



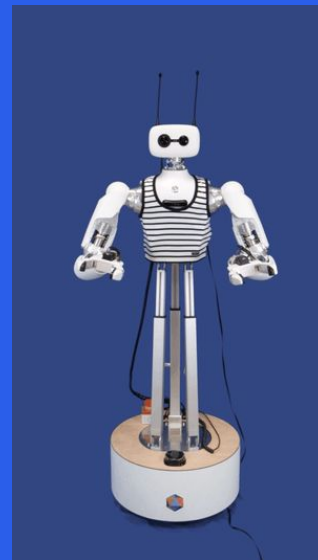
PEFT



smolagents

What's Hugging Face? 🤗

We also do robotics!



Open vs Closed



	Open	Closed / Proprietary
Security	Models can be self-hosted , data stays in your environment	Models cannot be self-hosted. Data is sent outside your environment to vendor
Control	The timing and nature of updates are controlled by you	Updates and changes to performance can happen without notice
Customization	Full source code access to customize the model for your needs	Limited ability to customize for your needs
Transparency	Inspect code and data provides better auditability and understandability	No ability to audit or understand performance
Cost	Typical lower long term cost due to smaller model size	larger model size and proprietary premium often balanced by decreased cost from server-side optimization
Latency	Lower latency due to on premise and smaller model sizes	Often greater latency due to larger model sizes + API latency
Quality	No single approach is best. Each use case will vary. Proprietary is typically closer to the frontier of performance .	
Examples	 OpenAI  Meta  deepseek  MISTRAL AI  Microsoft	 OpenAI ANTHROPIC

How HF democratizes ML/AI



ML engineers who
train models from
scratch

Software engineers
who use inference
APIs, ML engineers
who fine-tune
models

Anyone who
can use a GUI



Datasets



Models



Spaces



Used everywhere in the AI world

15,000+ startups and enterprises



Open source contributors



**Hugging
Face**

Cloud partners



Hardware partners



On-prem partners



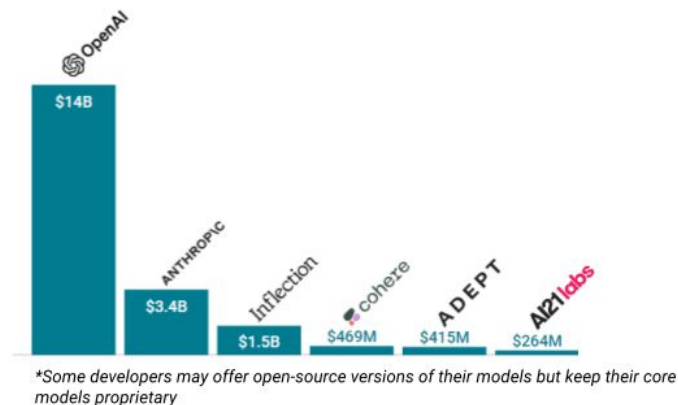
Is the market interested?



The private market is split into open vs. closed

Disclosed equity funding to LLM developers (as of 10/27/2023)

Closed-source LLMs



Open-source LLMs



*Excludes open-source developers that have not raised equity funding

Is this actually possible?

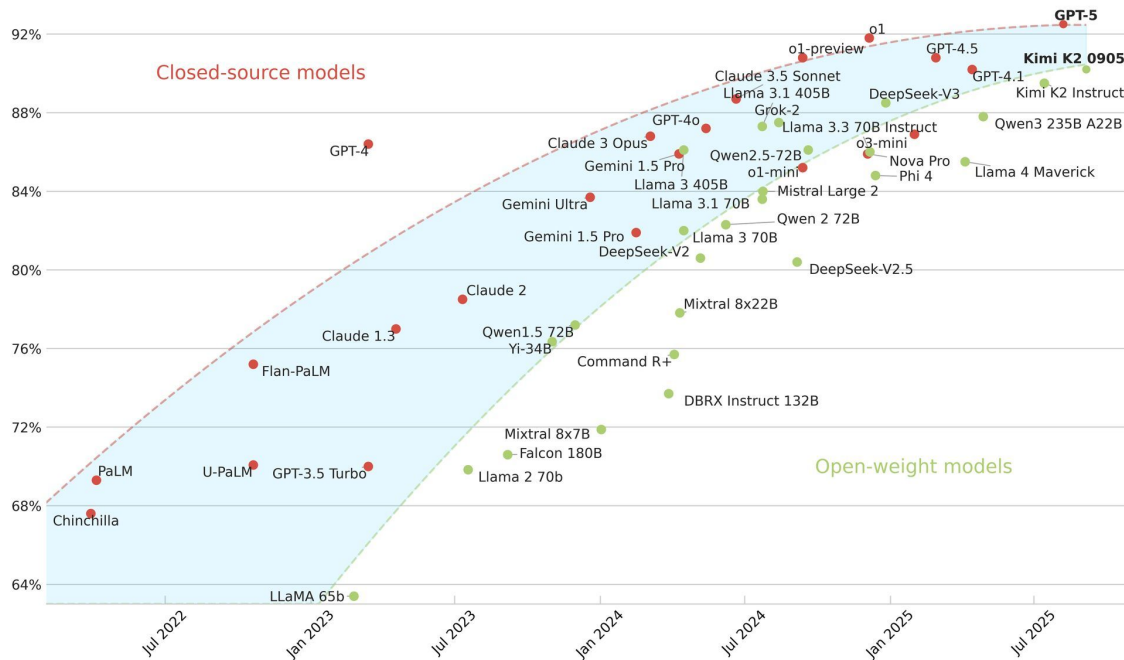


Closed-source vs. open-weight models

@maximelabonne

Chinese MoE models now dominate frontier open-source releases.

MMLU (5-shot)



The role of a ML Engineer at HF

It depends on your specific team:

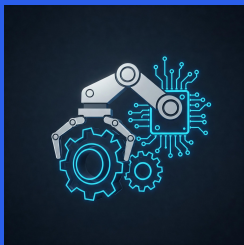
- Transformers (or any particular project)
- Research
- Robotics
- Advocacy

...



What is LeRobot?

The platform which provides models, datasets, tools for real-world robotics in PyTorch, lowering the barrier to entry to robotics, and encouraging to share datasets and pretrained models.



Policies



Datasets



Tools



Policies

Supports SoTA models in robotics

Enables easy
training/inference
with imitation
learning, RL, VLA
models





Datasets

> 22K

Datasets
shared on
the HF Hub

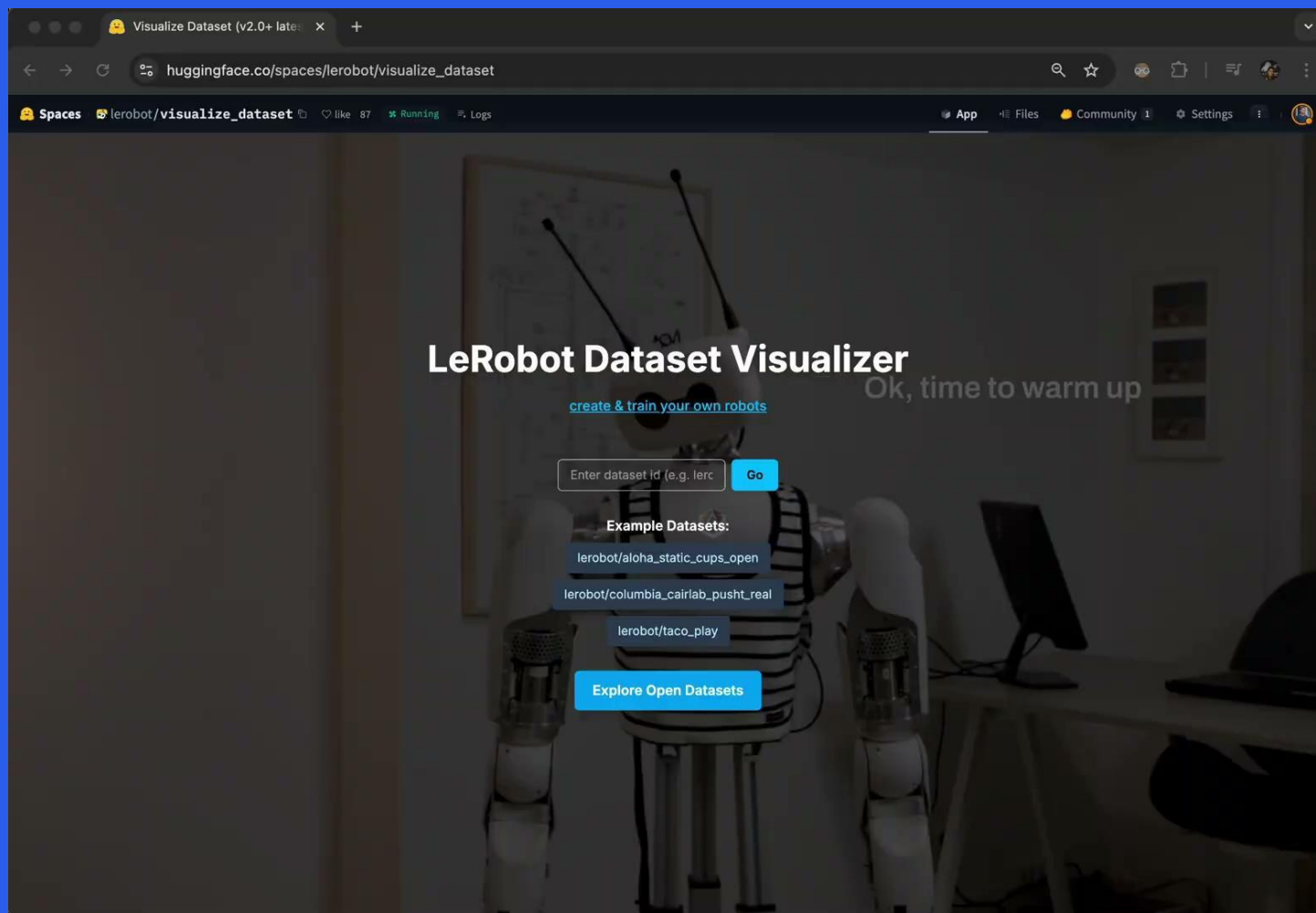
The screenshot shows the Hugging Face Datasets page. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Community, Docs, Pricing, and a user profile. The main content area is divided into a left sidebar and a main grid. The sidebar has tabs for Main (selected), Tasks, Libraries, Languages, Licenses, and Other. Under Tasks, there are filters for Filter Tasks by name and Reset Tasks. The sidebar also lists various task categories: Multimodal (Image-Text-to-Text, Visual Question Answering, Video-Text-to-Text, Visual Document Retrieval, Any-to-Any), Computer Vision (Depth Estimation, Image Classification, Object Detection, Image Segmentation, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Unconditional Image Generation, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D, Image-to-3D, Image Feature Extraction), and Natural Language Processing (Text Classification, Text-to-Text, Text-to-Speech, Speech-to-Text, Text-to-Audio, Audio-to-Text, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D, Image-to-3D, Image Feature Extraction). The main grid displays a list of datasets with filters for name, full-text search, and sort order (Most likes). The datasets shown include:

- agibot-world/AgiBotWorld-Alpha**: Updated Sep 29, 49.8M, 11.7k, 202
- nvidia/PhysicalAI-Robotics-GR00T-X-Embodi...**: Updated Sep 17, 1.95M, 163
- jxu124/OpenX-Embodiment**: Updated Oct 16, 2024, 10.7k, 80
- InternRobotics/OmniWorld**: Updated 20 days ago, 5.54B, 44.7k, 70
- lvhaidong/LAFAN1_Retargeting_Dataset**: Updated Apr 6, 1.45k, 51
- physical-intelligence/libero**: Updated Feb 2, 273k, 17.5k, 42
- agibot-world/AgiBotWorld-Beta**: Updated 22 days ago, 248M, 19.3k, 42
- yaak-ai/L2D**: Updated Sep 29, 19M, 9.35k, 41
- omniiretarget/OmniRetarget_Dataset**: Updated 23 days ago, 2.51k, 32
- InternRobotics/MotionMillion**: Updated Sep 19, 1.25M, 877, 29
- InternRobotics/InternScenes**: Updated Sep 19, 10.7k, 29
- InternRobotics/InternData-N1**: Updated 5 days ago, 38.6k, 29
- nvidia/PhysicalAI-SimReady-Warehouse-01**: Updated May 16, 753, 2.21k, 26
- InternRobotics/InternData-A1**: Updated Sep 19, 8.62M, 772, 26
- x-humanoid-robomind/RoboMIND**: Updated 1 day ago, 17.6k, 25
- lerobot/pusht**: Updated Sep 27, 25.7k, 5.7k, 24
- nvidia/PhysicalAI-Robotics-GraspGen**: Updated Jun 21, 25.5k, 465, 24
- nvidia/PhysicalAI-Robotics-NuRec**: Updated 4 days ago, 572, 22



Tools

Provides tools
for dataset
upload,
visualization,
and replay



SmolVLA

What is it? And how to build it?



Scan to read more



Inputs

Text



The task description containing the robot's objective

Example: Pick the yellow lego block and put it in the box

Vision



Video frames containing multiple viewpoints of the joint positions

States



Captures robots' current physical configuration: Joint positions, end-effector pose, gripper status

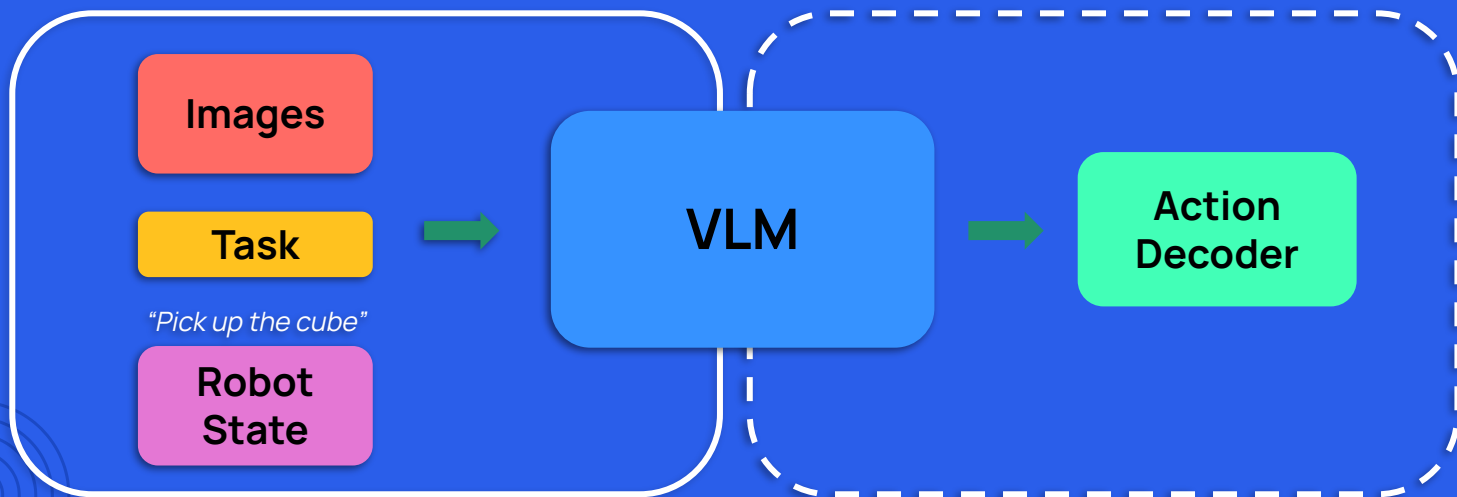
Actions



Target robot action, representing what robot should do next (e.g., 6D pose or joint angles)

What is VLA?

Vision-Language-Action models learn to generate actions based on visual input and natural language commands.



SmolVLA performance

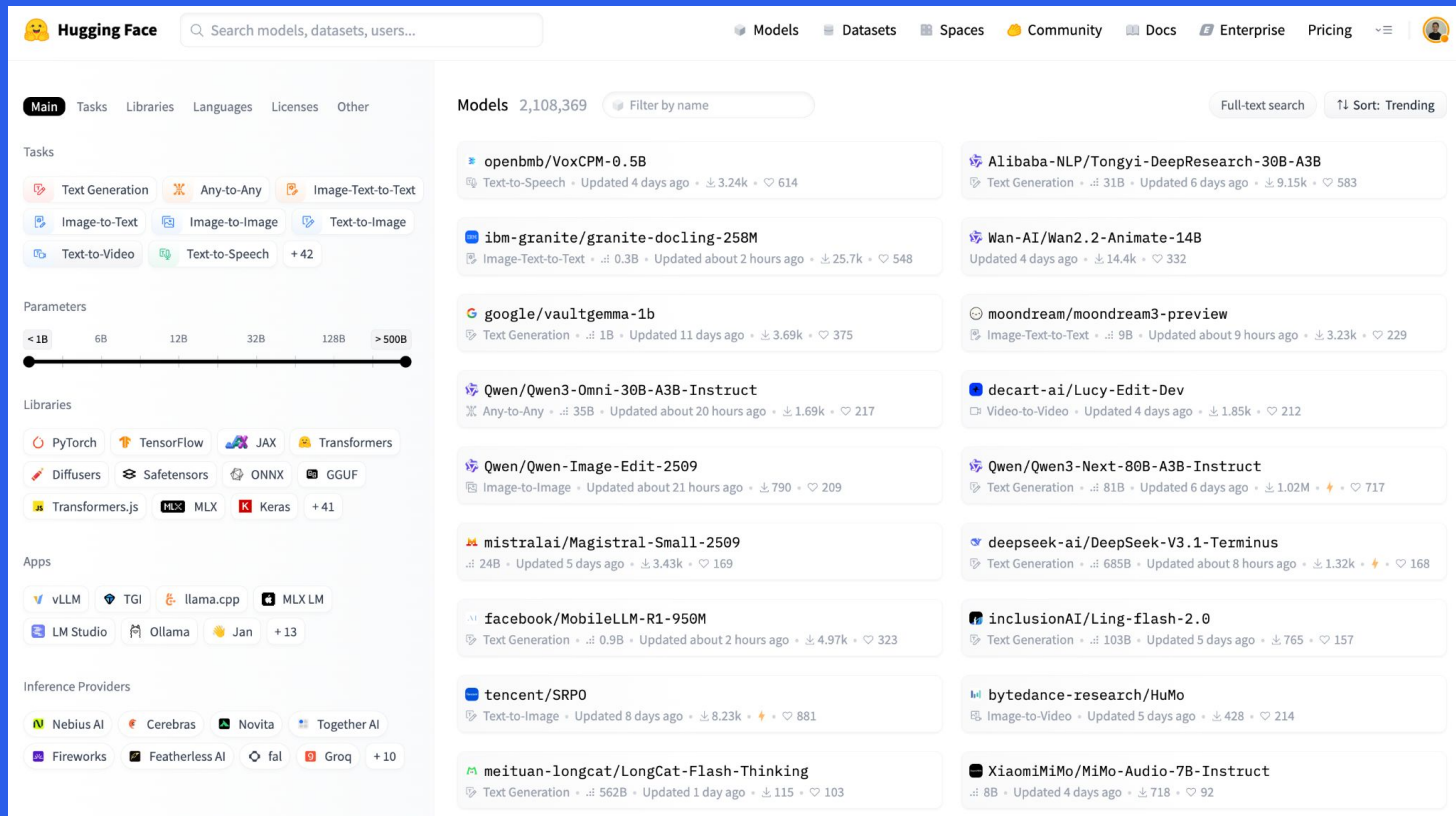


How do we build with it?

Find model → Customize → Deploy



Find the model



The screenshot shows the Hugging Face website's 'Models' section. The left sidebar contains navigation links for 'Main', 'Tasks', 'Libraries', 'Languages', 'Licenses', and 'Other'. Under 'Tasks', there are buttons for 'Text Generation', 'Any-to-Any', 'Image-Text-to-Text', 'Image-to-Text', 'Image-to-Image', 'Text-to-Image', 'Text-to-Video', 'Text-to-Speech', and '+42'. The 'Parameters' section features a slider from '<1B' to '>500B'. The 'Libraries' section lists various frameworks like PyTorch, TensorFlow, JAX, Transformers, etc. The 'Apps' section shows vLLM, TGI, llama.cpp, etc. The 'Inference Providers' section lists Nebius AI, Cerebras, etc. The main content area displays a list of models with filters and sorting options. The models listed include openbmb/VoxCPM-0.5B, Alibaba-NLP/Tongyi-DeepResearch-30B-A3B, ibm-granite/granite-docling-258M, Wan-AI/Wan2.2-Animate-14B, google/vaultgemma-1b, moondream/moondream3-preview, Qwen/Qwen3-Omni-30B-A3B-Instruct, decart-ai/Lucy-Edit-Dev, Qwen/Qwen-Image-Edit-2509, Qwen/Qwen3-Next-80B-A3B-Instruct, mistralai/Magistral-Small-2509, deepseek-ai/DeepSeek-V3.1-Terminus, facebook/MobileLLM-R1-950M, inclusionAI/Ling-flash-2.0, tencent/SRPO, bytedance-research/HuMo, meituan-longcat/LongCat-Flash-Thinking, and XiaomiMiMo/MiMo-Audio-7B-Instruct. Each model entry shows its name, task, update time, size, and popularity metrics.

Hugging Face Search models, datasets, users...

Models Datasets Spaces Community Docs Enterprise Pricing

Main Tasks Libraries Languages Licenses Other

Tasks

Text Generation Any-to-Any Image-Text-to-Text

Image-to-Text Image-to-Image Text-to-Image

Text-to-Video Text-to-Speech +42

Parameters

<1B 6B 12B 32B 128B >500B

Libraries

PyTorch TensorFlow JAX Transformers

Diffusers Safetensors ONNX GGUF

Transformers.js MLX Keras +41

Apps

vLLM TGI llama.cpp MLX LM

LM Studio Ollama Jan +13

Inference Providers

Nebius AI Cerebras Novita Together AI

Fireworks Featherless AI fal Groq +10

Models 2,108,369 Filter by name

Full-text search Sort: Trending

openbmb/VoxCPM-0.5B

Text-to-Speech • Updated 4 days ago • 3.24k • 614

Alibaba-NLP/Tongyi-DeepResearch-30B-A3B

Text Generation • 31B • Updated 6 days ago • 9.15k • 583

ibm-granite/granite-docling-258M

Image-Text-to-Text • 0.3B • Updated about 2 hours ago • 25.7k • 548

Wan-AI/Wan2.2-Animate-14B

Updated 4 days ago • 14.4k • 332

google/vaultgemma-1b

Text Generation • 1B • Updated 11 days ago • 3.69k • 375

moondream/moondream3-preview

Image-Text-to-Text • 9B • Updated about 9 hours ago • 3.23k • 229

Qwen/Qwen3-Omni-30B-A3B-Instruct

Any-to-Any • 35B • Updated about 20 hours ago • 1.69k • 217

decart-ai/Lucy-Edit-Dev

Video-to-Video • Updated 4 days ago • 1.85k • 12

Qwen/Qwen-Image-Edit-2509

Image-to-Image • Updated about 21 hours ago • 790 • 209

Qwen/Qwen3-Next-80B-A3B-Instruct

Text Generation • 81B • Updated 6 days ago • 1.02M • 717

mistralai/Magistral-Small-2509

24B • Updated 5 days ago • 3.43k • 169

deepseek-ai/DeepSeek-V3.1-Terminus

Text Generation • 685B • Updated about 8 hours ago • 1.32k • 168

facebook/MobileLLM-R1-950M

Text Generation • 0.9B • Updated about 2 hours ago • 4.97k • 323

inclusionAI/Ling-flash-2.0

Text Generation • 103B • Updated 5 days ago • 765 • 157

tencent/SRPO

Text-to-Image • Updated 8 days ago • 8.23k • 881

bytedance-research/HuMo

Image-to-Video • Updated 5 days ago • 428 • 214

meituan-longcat/LongCat-Flash-Thinking

Text Generation • 562B • Updated 1 day ago • 115 • 103

XiaomiMiMo/MiMo-Audio-7B-Instruct

8B • Updated 4 days ago • 718 • 92

<https://huggingface.co/models>

Use it with 🧡 Transformers

The Transformers Library: standardizing model definitions

Published May 15, 2025

[Update on GitHub](#)



Lysandre
lysandre

Follow



Arthur Zucker
ArthurZ

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Pedro Cuenca
pcuenca

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Julien Chaumond
julien-c

Following

TLDR: Going forward, we're aiming for Transformers to be the pivot across frameworks: if a model architecture is supported by transformers, you can expect it to be supported in the rest of the ecosystem.

Transformers was created in 2019, shortly following the release of the BERT Transformer model. Since then, we've continuously aimed to add state-of-the-art architectures, initially focused on NLP, then growing to Audio and computer vision. Today, transformers is the default library for LLMs and VLMs in the Python ecosystem.



<https://huggingface.co/blog/transformers-model-definition>



Use it with 🙌 Transformers

```
from transformers import pipeline

# Load the text generation pipeline with Qwen3
pipe = pipeline(
    "text-generation",
    model="Qwen/Qwen3-0.6B",
    dtype="auto",
    device_map="auto"
)

# Define the conversation (messages)
messages = [
    {
        "role": "user",
        "content": "Give me a short introduction to large language model."
    }
]

# Generate the response
outputs = pipe(
    messages,
    max_new_tokens=512,
)

# The pipeline returns structured output
print(outputs[0]["generated_text"][-1]["content"])
...
<think>
Okay, the user wants a short introduction to a large language model. Let me start by ...
</think>

A large language model (LLM) is a type of artificial intelligence that can understand, generate, and respond to human language. These models are trained on massive datasets to understand complex topics and produce creative, context-aware content. They are used in various applications, from writing essays to providing translations and answering questions.
...
```

<https://huggingface.co/docs/transformers/index>



Use it with 🤖 Transformers

```
from transformers import AutoProcessor, AutoModelForCausalLM
import torch

model_name = "Qwen/Qwen3-0.6B"
processor = AutoProcessor.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(
    model_name,
    dtype="auto",
).to("cuda")

messages = [
    {
        "role": "user",
        "content": [
            {"type": "text", "text": "Give me a short introduction to large language model"},
        ]
    },
]

inputs = processor.apply_chat_template(
    messages,
    add_generation_prompt=True,
    tokenize=True,
    return_dict=True,
    return_tensors="pt",
).to(model.device)

generated_ids = model.generate(**inputs, do_sample=False, max_new_tokens=256)
generated_texts = processor.batch_decode(
    generated_ids,
    skip_special_tokens=True,
)

print(generated_texts[0])
...
<think>
Okay, the user wants a short introduction to a large language model. Let me start by ...
</think>
```

A large language model (LLM) is a type of artificial intelligence that can understand, generate, and respond to human language. These models are trained on massive datasets to understand complex topics and produce creative, context-aware content. They are used in various applications, from writing essays to providing translations and answering questions.



Customize with **TRL**

TRL allows you to fine-tune VLMs using different methods:
SFT, DPO, MPO, GRPO, GSPO, RLOO, and Online DPO.
PEFT and quantization support

```
from trl import SFTTrainer
from datasets import load_dataset

trainer = SFTTrainer(
    model="Qwen/Qwen3-0.6B",
    train_dataset=load_dataset("trl-lib/Capybara", split="train"),
)
trainer.train()
```

Deploy model



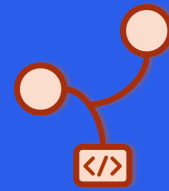
Inference and serving frameworks

Inference Providers (TGI)

Hugging Face Endpoints 

 VLLM

LLaMA 



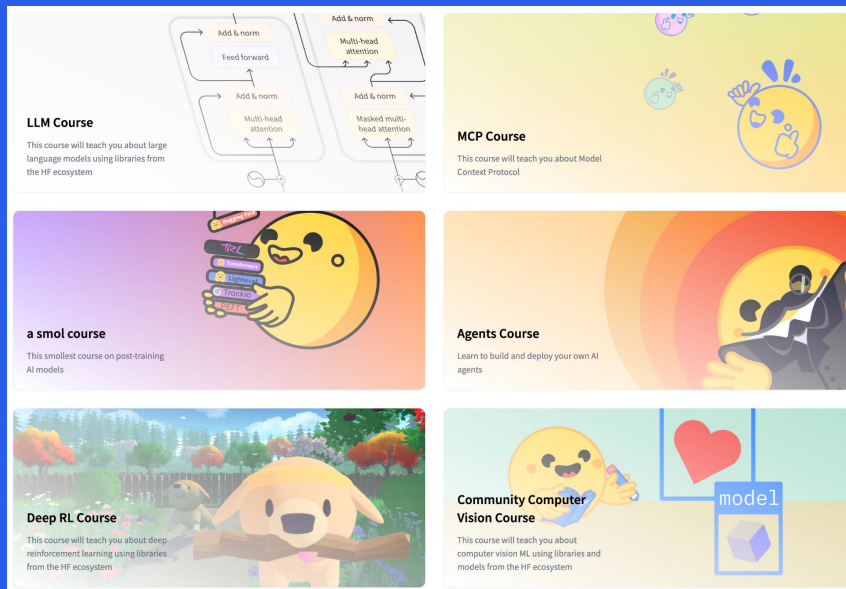
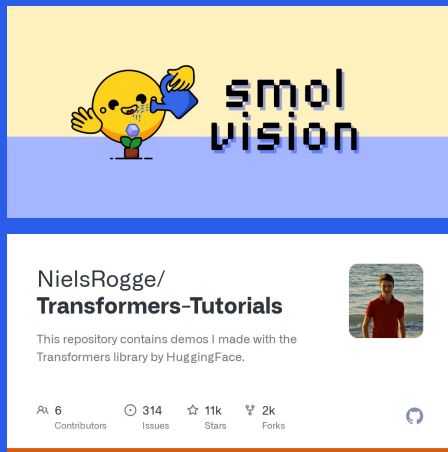
SG L



Resources



- hf.co/blog
- hf.co/learn
- [Transformers-Tutorials repo](#)
- [smol-vision repo](#)



Thanks!

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